Assignment-1

Section-A

1. True
2. True
3. False
4. True
5. True
6. False
7. False
8. True
9. True
10. True

Section-B

1.

Model	Loss Function	Regularizer
SVM	→ Hinge Loss	$L_2 \|\omega^2\|$
LASSO	→ Mean Square Error	$L_1 \|\omega_1\|$
RIDGE	→ Mean Square Error	$L_2 \|\omega^2\|$

3. (a) Loss Functions that are continous and differentiable can be optized using gradient descent such as:
 • Squared loss • Logistic Loss • Exponential Loss.
 This is because gradient descent required the gradient to exist.

 (b) Loss functions that are twice differentiable can be optized using Newton's method such as • Squared loss
 • Logistic loss

 Newton's method requires second-order derivatives, which non-smooth losses like Hinge loss or 0-1 loss do not have.

Section-C

1. Underfitting occurs when the model is too simple to capture the underlying pattern in the data, leading to high bias.

2. It implies that the model has high bias or the data may be very noisy or not informative enough, so the model cannot learn the true relationship.

3. Bagging trains multiple models on different bootstrap samples and averages their predictions, which reduces sensitivity to individual data points and lower variance.

4. Boosting mainly reduces bias by focusing on hard-to-classify examples but it can increase variance if the model becomes too complex.

Section-D

1. • Use efficient data structure (Ex- KD-tree, Ball Tree)
   • Reduce dataset size (sampling or prototype selection)
   • Reduce dimensionality (PCA, feature selection)
   • Use approx. nearest neighbours.

2. (a) No. Squaring the Euclidean distance does not change the order of distances, So, the nearest neighbours remain the same.

   (b) No, Squaring Distance does not fix the curse of dimensionality, since relative distances still became similar in high dimensions.

3. In high dimensions, data becomes sparse and distances between points became similar, making it hard to identify meaningful nearest neighbors.

4. Small K → low bias, high variance
   Large K → high bias, low variance.

5. KNN is preferred when –
   • The decision boundary is highly non-linear.
   • there is enough data and low dimensionality
   • No strong parametric assumptions are desired.

Section- E

1. At a leaf, we want to minimize :- $\sum_{i=1}^{n}(y_i-c)^2$

   Take derivative w.r.t. c

$$\frac{d}{dc}\sum(y_i-c)^2 = -2\sum(y_i-c) = 0$$

$$c = \frac{1}{n}\sum y_i$$

   So, the optimal prediction is the mean of target values.

2. Minimum Gini :- 0 (when all samples belong to one class)

   Maximum Gini :- $1-\sum_{i=1}^{3}\left(\frac{1}{3}\right)^2 = 1-\frac{1}{3} = \frac{2}{3}$

3. Decision Trees are myopic because they make greedy, local decisions at each split without considering future splits or global optimality.

4. (i) Pre-pruning → Limit tree growth using max-depth, min. samples per leaf etc.

(ii) Post-pruning → Grow a full tree and then remove branches that do not improve validation performance.

→ Section-F

1. Yes, using out-of-bag (OOB) samples. Each tree is trained on a bootstrap sample, leaving out ~~leaving~~ about 36% of data, which can be used for testing.

2. (1) Bagging → Trains models independently on random samples to reduce variance.

(2) Boosting → Trains models sequentially, focusing more on previously misclassified points to reduce bias.